Check for updates

REVIEW

Cardiac Arrest and Neurologic Recovery: Insights from the Case of Mr. Damar Hamlin

Romergryko G. Geocadin, MD, FNCS, FAAN, FANA,[1] Sachin Agarwal, MD, MPH,[2] Adeline L. Goss, MD,[3] Clifton W. Callaway, MD, PhD,[4] and Megan Richie, MD[5]

The association between brain injury after cardiac arrest and poor survival outcomes has led to longstanding pessimism. However, the publicly witnessed cardiac arrest, resuscitation, and acute management of Mr. Damar Hamlin and his favorable neurologic recovery provides some optimism. Mr. Hamlin's case highlights the neurologic advances of the last 2 decades and presents the opportunity to improve outcomes for all cardiac arrest patients in key areas: (1) effectively implementing the American Heart Association "Chain of Survival" to prevent initial brain injury and promote neuroprotection; (2) revisiting the process of neurologic prognostication and re-defining the brain recovery during the early periods, and (3) incorporating neurorehabilitation into existing cardiac rehabilitation models to support holistic recovery.

ANN NEUROL 2023;00:1–6

A Lifesaving Opportunity Arising from Monday Night Football Tragedy

Millions of football fans worldwide were shaken on January 2, 2023, when 24-year-old Damar Hamlin, playing at the safety position for the Buffalo Bills of the National Football League, suffered an out-of-hospital cardiac arrest (OHCA) after a tackle during a Monday Night Football game. After a successful, well-orchestrated resuscitation on the field, he was transferred in a very critical condition to the University of Cincinnati Medical Center for post-OHCA care.[1] This was followed by a period of uncertainty, commonly encountered by neurologists, cardiologists, and intensivists, as the world wondered to what extent Mr. Hamlin's brain would recover. But remarkably, just 4 days later, he manifested early signs of neurological recovery and was able to communicate in writing. He was liberated from the ventilator the following day and discharged home within nine days of his OHCA.

This very public event served as an excellent opportunity to remind key stakeholders, including patients with OHCA, their families, health care professionals, resuscitation researchers, and community-based advocacy groups that there can be optimism about the recovery from cardiac arrest, despite much historical pessimism. The fantastic recovery provides insights into some of the critical aspects of resuscitation, advances in critical care, and promising future research directions that allow outcomes like his for patients with OHCA everywhere. The key areas with neurologic importance that can be highlighted here are: (1) effectively implementing the American Heart

View this article online at wileyonlinelibrary.com. DOI: 10.1002/ana.26619

Received Feb 9, 2023, and in revised form Feb 23, 2023. Accepted for publication Feb 23, 2023.

Address correspondence to Dr Geocadin, The Johns Hopkins University School of Medicine, 600 N. Wolfe Street, Phipps 455 Baltimore, MD 21287 USA. E-mail: rgeocad1@jhmi.edu

Sachin Agarwal, Adeline L. Goss, Clifton W. Callaway, Megan Richie these authors contributed equally to this work.

From the [1]Departments of Neurology, Anesthesiology-Critical Care Medicine and Neurosurgery, Johns Hopkins University School of Medicine, Baltimore, MD, United States; [2]Department of Neurology (Neurocritical Care), Columbia University Irving Medical Center, New York, NY, United States; [3]Department of Internal Medicine, Highland Hospital, Oakland, CA, United States; [4]Department of Emergency Medicine, University of Pittsburgh School of Medicine, Pittsburgh, PA, United States; and [5]Department of Neurology, University of California – San Francisco School of Medicine, San Francisco, CA, United States

© 2023 The Authors. *Annals of Neurology* published by Wiley Periodicals LLC on behalf of American Neurological Association.
This is an open access article under the terms of the Creative Commons Attribution-NonCommercial License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited and is not used for commercial purposes.

Association (AHA) "Chain of Survival" to prevent initial brain injury and promote neuroprotection, (2) revisiting the process of neurologic prognostication and re-defining the recovery process of the brain in the early periods, and (3) incorporating neurorehabilitation in existing cardiac rehabilitation models for holistic recovery.

Implementing the Chain of Survival: Closing the Gap between Clinical Trial and Real-World Outcomes

There has been pessimism associated with brain injury after OHCA because of overall poor survival outcomes, which, at least for a decade, have lingered around 10% of persons for whom cardiopulmonary resuscitation (CPR) is started in the United States.[2] Although these real-world statistics are troubling, OHCA clinical trials investigating the benefit of targeted temperature management (TTM) in the early 2000s, with survival in 49% to 59% of patients with OHCA receiving TTM and 32% to 45% in the control group,[3,4] completely transformed the approach to patients who reach the hospital after CPR. These trials provided optimism and challenged us to redouble efforts in both prehospital and in-hospital care for our diverse US populations. More recent TTM trials reported 6-month survival rates of 48% to 52% in both control and intervention arms indicating how much critical care outcomes have improved.[5,6] However, improving survival requires more than better critical care. Mr. Hamlin's very public resuscitation illustrates how all the steps in the chain of survival and recovery can be in place with proper training and preparation. This event provides us with a roadmap and motivation to re-examine how to make the best care available for all patients with OHCA.

Revisiting the "Chain of Survival" framework[7] (Fig) may guide our introspection into Mr. Hamlin's remarkable recovery. This framework was developed to undertake prompt resuscitation and promote survival for patients with an OHCA. As with other critical care illnesses, the extent of brain injury caused by lack of blood flow to the brain during OHCA (no-flow) is a significant determinant of the quality of life after survival.[8] The highest priority to minimize neuronal injury after OHCA is to minimize the no-flow duration by rapidly establishing a return of spontaneous circulation (ROSC).[7] High-quality and immediate CPR with chest compression, including the use of advance CPR assist devices, can reduce brain injury while in a low-flow state. Immediately administering electrical defibrillation when appropriate can achieve ROSC in the shortest possible time. As reported and witnessed in Mr. Hamlin's case, all the steps of this Chain of Survival, including rapid recognition and activation of emergency response, were rapidly deployed on the field.

Neurologists and neurocritical care specialists can provide important contributions early in the Chain of Survival. Even when the primary etiology of OHCA is cardiac, potential or evident brain injury must be addressed simultaneously. As provided in the new reports, Mr. Hamlin was transferred to the University of Cincinnati Medical Center, where he received state-of-the-art post-OHCA critical care, including TTM, to minimize further brain injury.[1]

Although the role of the neurologist has historically focused on the prognostication of the neurologic outcome, neurologists need to work with other healthcare providers as early as possible in the post-resuscitation period to help promote brain recovery by optimizing systemic factors (blood pressure, oxygen levels, temperature, and acid–base

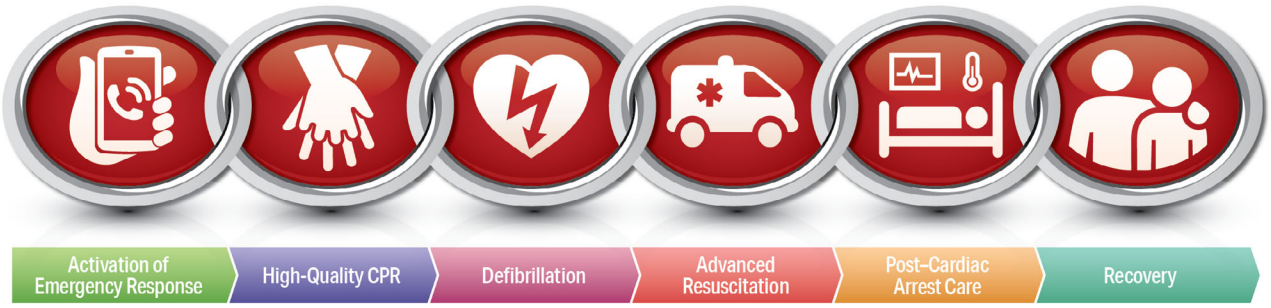

Figure: The updated AHA Chain of Survival. The Chain of Survival includes elements of the following: Prevention and preparedness, including responder training, early recognition of cardiac arrest, and rapid response. Activation of the emergency response system, either outside of or within the hospital. High-quality CPR and early defibrillation of ventricular fibrillation and pulseless ventricular tachycardia heart rhythms. Advanced resuscitation interventions, including medications, advanced airway interventions, and extracorporeal CPR. Post-cardiac arrest care, including critical care interventions and optimal targeted temperature management. Recovery, including effective rehabilitative support and early screening for physical, cognitive, emotional, and family needs. AHA = American Heart Association; CPR = cardiopulmonary resuscitation; OHCA = out-of-hospital cardiac arrest. (Reprinted with permission. Circulation. 2020; 142:S580-S604 @2020 American Heart Association, Inc.)

15318249, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ana.26619, Wiley Online Library on [24/04/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

balance),[9] as well as anticipate and manage neurologic complications, such as disorders of consciousness (especially coma and stupor), seizures, and brain swelling in the intensive care unit.[10]

The introduction of TTM 2 decades ago led to major advancements in post cardiac arrest care. The first two randomized controlled trials (RCTs) found that targeting 32°C to 34°C was superior to normothermia,[3,4] whereas subsequent RCTs found TTM to 33°C or 36°C or even fever avoidance (< 37.8°C) resulted in similar survival and favorable outcome rates.[5,6] All these trials reported better outcomes than other RCTs during the pre-TTM era.[11] The implementation of TTM protocols was not limited to temperature control alone but also dictated aggressive critical care to maintain physiologic goals and minimize premature withdrawal of life-sustaining therapies (WLSTs).[12,13] These concurrent interventions may have been central to improving outcomes.[12,13] In this heterogeneous OHCA population, identification of one temperature target is challenging; for example, some large case series[14,15] suggest benefit of 33°C as the target temperature, whereas a meta-analysis favored 36°C or < 37.8°C[16]; and the American Academy of Neurology (AAN) practice guidelines recommend either 33°C or 36°C targets, although slightly favoring 33°C.[11] With the similar neuroprotective effects of targeting 33°C, 36°C, or < 37.8°C core temperatures in the RCTs, the decision for the temperature target is at the discretion of the treating team, with consideration of individual patient factors.[13]

Revisiting the Process of Neurologic Prognostication and re-Defining the Recovery Process of the Brain in the Early Periods

Over the years, we, as a field of resuscitation science, have realized that our ability to provide neurologic prognostication for comatose patients in the intensive care unit is subject to many confounders, such as body temperature, sedatives, and associated systemic complications and comorbidities that tend to make the neurologic status appear worse. The neuro-prognostication-based studies and tools developed have almost entirely focused on the prediction of poor outcomes. Recently, we have recognized a shift with the development of electrophysiologic, imaging, and serum biomarkers that can predict good outcomes, estimate brain injury severity, and track recovery.[17–19]

Historically, neurological prognostication in comatose cardiac arrest survivors was dominated by an investigation published close to 4 decades ago.[20] In absence of any effective neuroprotective interventions available at that time, in a cohort of patients with overall 13% good or moderately disabled outcomes, the neuroprognostication scheme focused on identifying patients with poor outcomes and relied heavily on clinical neurologic findings at specific timeframes (ie, days 1, 3, and 7) after hypoxia or ischemia.[20] Multiple neurologic prognostication studies over the following years followed and culminated in the 2006 practice guideline from the AAN on the prediction of poor outcomes in comatose survivors of cardiac arrest.[21] Despite the AAN guideline on the use of TTM as an effective neuroprotective therapy for OHCA,[11] the AAN neurologic prognostication guideline has not been updated, resulting in many reports of inappropriate pessimistic prognostication in the first 3 to 5 days for many patients and especially those treated with TTM. These observations were included in a scientific statement that revisited the practice of neurologic prognostication in 2019.[22] After an extensive review, the multidisciplinary panel formed by the AHA concluded that the overwhelming studies published were of low quality, and, as a result, the confidence in the predictors was also low. Further, the confidence intervals around point estimates for predicted recovery were wide.[22,23] Thus, this historical approach based on purely clinical examination findings over a few days is discouraged from routine care.

In parallel with the development of novel therapies with the potential to mitigate neurologic injury, the field of neurologic prognostication has developed many new tools. Although no single tool perfectly predicts the outcome of coma after OHCA, a modern approach to prognostication uses multiple modalities to assess signs of brain injury, including neurophysiology, imaging, and biochemical markers.[22] These multimodality biomarkers can detect not only early injury but also how the brain responds to interventions.[19] These tools can guide treating teams to adjust or increase therapeutic intensity in the early periods rather than just reporting signs of poor response that might prompt families or clinical teams to WLST. Providers with experience in selecting and interpreting these brain biomarkers should be involved with these high-stakes evaluations.

The 2019 AHA panel evaluating the state of neurologic prognostication identified self-fulfilling prophecies as among the most important biases in this area.[22] Almost 90% of neuroprognostication studies showed a high risk for self-fulfilling prophecies in clinical practice.[24] Most studies do not characterize the cause of death as brain death, cardiac death, or death after WLST. Furthermore, studies that did not report blinding of treating teams show that as much as 59% to 80% of deaths were preceded by WLST.[23] As self-fulfilling prophecy is a significant confounder and cause of death, clinicians must be mindful of

15318249, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ana.26619, Wiley Online Library on [24/04/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

the use of any single prognostic parameter and instead, rely on the guideline-recommended multimodal prognostication schemes.[9]

Whereas historical guidelines pre-TTM suggested the earliest time for neurological prognostication should be at 72 hours after ROSC,[9,24] there is increasing evidence that more extended observation periods (> 72 hours) reveal more patients who awaken from coma with favorable outcomes.[25,26] The 2 TTM trials protocolized neuroprognostication to be undertaken after 96 hours only[5,6] and the 2019 AHA panel recommended that assessing predictors of neurological outcome should extend the observation period to 7 days after either the end of TTM or the suspension of sedation (whichever occurs later).[22] A recent study showed that a delay of neurologic prognostication beyond 7 days resulted in about 10% more patients with OHCA with favorable outcomes.[27] As we reflect on the time frame of 3 days when neurologic assessments were done in the Levy study in 1985,[20] there is optimism that favorable outcomes still occur when we extend the observation periods beyond 3 to 7 days for neurologic prognostication.[27]

Considering the limitations of neurologic prognostication studies, a pragmatic approach to neurologic prognostication would be to extend the period of observation to 7 days, especially after TTM or suspension of sedation (whichever occurs later)[22] prior to predicting poor outcome and deciding on WLST. Exceptions to this approach include patients with overt medically refractory cardiovascular failure, intractable multi-organ failure, or those that satisfy the diagnosis of death by neurologic criteria.[28] As advised by some practice guidelines that also acknowledged uncertainty of existing evidence,[9,24] neurological prognostication should only be undertaken using multiple modalities. Although the neurologic examination in the absence of confounders can identify neurologically devastated patients (ie, persistent unresponsiveness, absent pupillary light reflex, and/or corneal reflex) and is the core prognostic parameter, it should be supported with other modalities that can help also predict unfavorable recovery, such as bilateral absence of N20 wave on somatosensory evoked potential testing, very low amplitude and nonreactive electroencephalogram (EEG), and extensive abnormality on Diffusion-Weighted Magnetic Resonance Imaging.[9,24]

Long-Term Care and Rehabilitation of Mr. Hamlin and OHCA Survivors

The science of recovery and OHCA survivorship so far has used rather coarse functional scales to characterize neurological outcomes.[29] By these scales, Mr. Hamlin appears to be someone who is "neurologically intact" or who has achieved an excellent functional outcome. Unfortunately, survivors of OHCA, like many survivors of critical illness, often suffer from a spectrum of physical, cognitive, behavioral, and social issues.[30] These noncardiac sequelae are highly prevalent and persistent for months and years after OHCA. Based on limited existing literature, the most common neurological problems include cognitive impairments, musculoskeletal dysfunction (including spasticity), movement disorders (including myoclonus), and seizures. Patients with more severe brain injury may have persistent disorders of consciousness (including minimally conscious and unresponsive wakeful states). Even among apparently "neurologically intact" patients, the extra effort to adapt to subtle deficits can lead to daily fatigue, which is the most common complaint from surviving patients.[31] Neuropsychologic problems may not be readily evident and may require testing for cognitive impairment with executive function, attention, memory, language, and visuospatial perception.[30] Healthcare providers need to look for these issues and educate patients and their caregivers and help develop strategies to manage them.

Cardiac arrest survivors have long-term needs that are only now being systematically studied.[30] Research suggests that potentially modifiable factors that contribute to the quality of recovery include post-OHCA psychological distress, the intensity of rehabilitation services after hospital discharge, as well as the physical and social environments in which recovery takes place.[30] Most current treatment recommendations focus on preventing a recurrence of cardiac arrest.[30] Currently, there are no guidelines or recommendations for neurorehabilitation for OHCA survivors in the United States. It is not clear what proportion of OHCA survivors receive any rehabilitation services at all. Outpatient cardiac rehabilitation programs, for which OHCA survivors may be currently eligible, primarily focus on improving exercise capacity. However, these programs can provide attention to other needs if they are recognized. Due to the high burden of functional and cognitive deficits and psychological distress associated with OHCA, rehabilitation strategies for OHCA survivors may warrant a more comprehensive and personalized approach where physical and occupational therapy, cognitive rehabilitation, and psychosocial care are integrated into the plan. This approach may promote engagement in exercise-based regimens and improve long-term functional recovery and quality of life.

The psychological experience of cardiac arrest is traumatic and life-altering, forcing many previously healthy individuals to suddenly confront an immediate existential threat for the first time.[32] One out of 3 OHCA

15318249, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ana.26619, Wiley Online Library on [24/04/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

survivors suffers from depression, anxiety, or post-traumatic stress.[32] Unlike other cardiovascular diseases, no clinical practice guidelines exist for assessing or treating psychological sequelae of OHCA. Due to the sudden, unexpected nature of the cardiac arrest event, caregivers also report stress-related responses, feelings of anxiety, and lower quality of life. Many strategies have been developed over the past century to help with psychological responses to trauma, and these tools may also benefit OHCA survivors. Currently, research is underway to characterize as well as to identify modifiable psychological targets to reduce secondary cardiovascular disease risk and improve the quality of life in OHCA survivors.[32] Even patients with "good outcomes" like Mr. Hamlin need to consider whether they or their families have long-term psychological needs that would benefit from self-management strategies, counseling, peer-support groups, or other therapies. Acknowledging these issues is especially important for restoring the well-being of individuals with pre-OHCA high-functioning status.

Conclusion

Whereas the overall survival after OHCA and particularly after a coma following OHCA, was historically low, the case of Mr. Damar Hamlin should inspire hope for patients and families that when the Chain of Survival is in place and executed well, favorable outcomes can be achieved. Much work is required to close the gap between the survival outcomes observed in clinical trials and the outcomes experienced by most communities. As neurologists, we can have a greater positive impact on patients with cardiac arrest by avoiding premature and overly pessimistic neurologic prognostication, supporting the early brain recovery process by avoiding and treating secondary brain injury, and helping to develop long-term brain-directed rehabilitation for meaningful recovery. For the health care providers and researchers in this area, this was a call to both implement what we know and address the many gaps where we need to know more.

Acknowledgments

None of the authors participated in or contributed to the medical care of Mr. Damar Hamlin. Dr. Geocadin participated in the development of guidelines and scientific statements related to cardiac arrest care from the AAN and AHA and receives support to his institution from the National Institutes of Health (NIH) to study neurologic recovery and prognostication and advance circulatory support after cardiac arrest (UG3 HL145269, R01 HL071568, R01 NS119825, and R01 HL155760). Dr. Agarwal receives support to his institution from the NIH to study cardiac arrest survivors (R01 HL153311). Dr. Callaway participated in the development of guidelines for cardiac arrest care from ILCOR and AHA and receives support to his institution from the NIH to study emergency care, including post-cardiac arrest care and prognostication (U24 NS100659, UH3 HL145269, R01 1NS119825, and R01 NS124642).

Author Contributions

R.G.G., A.G., and M.R. contributed to the conception, design, and drafting of the article. S.A. and C.W.C. contributed to the drafting of the article. All authors edited and approved the article.

Potential Conflicts of Interest

The authors declare that they have no conflict of interest.

References

1. https://www.youtube.com/watch?v=zcKuy6kP0a8. (Accessed February 7, 2023).
2. Tsao CW, Aday AW, Almarzooq ZI, et al. Heart disease and stroke Statistics-2022 update: a report from the American Heart Association. Circulation 2022;145:e153–e639.
3. Hypothermia after Cardiac Arrest Study Group. Mild therapeutic hypothermia to improve the neurologic outcome after cardiac arrest. N Engl J Med 2002;346:549–556.
4. Bernard SA, Gray TW, Buist MD, et al. Treatment of comatose survivors of out-of-hospital cardiac arrest with induced hypothermia. N Engl J Med. 2002;346:557–563.
5. Nielsen N, Wetterslev J, Cronberg T, et al. Targeted temperature management at 33°C versus 36°C after cardiac arrest. N Engl J Med. 2013;369:2197–2206.
6. Dankiewicz J, Cronberg T, Lilja G, et al. Hypothermia versus normothermia after out-of-hospital cardiac arrest. N Engl J Med. 2021;384: 2283–2294.
7. Berg KM, Cheng A, Panchal AR, et al. Part 7: Systems of Care: 2020 American Heart Association guidelines for cardiopulmonary resuscitation and emergency cardiovascular care. Circulation 2020;142: S580–s604.
8. Tasker RC, Menon DK. Critical care and the brain. JAMA 2016;315: 749–750.
9. Callaway CW, Donnino MW, Fink EL, et al. Part 8: post-cardiac arrest care: 2015 American Heart Association guidelines update for cardiopulmonary resuscitation and emergency cardiovascular care. Circulation 2015;132:S465–S482.
10. Mayasi Y, Geocadin RG. Updates on the Management of Neurologic Complications of post-cardiac arrest resuscitation. Semin Neurol 2021;41:388–397.
11. Geocadin RG, Wijdicks E, Armstrong MJ, et al. Practice guideline summary: reducing brain injury following cardiopulmonary resuscitation: report of the guideline development, dissemination, and implementation Subcommittee of the American Academy of neurology. Neurology 2017;88:2141–2149.
12. Rittenberger JC, Callaway CW. Temperature management and modern post-cardiac arrest care. N Engl J Med 2013;369:2262–2263.

15318249, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ana.26619, Wiley Online Library on [24/04/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

13. Morrison LJ, Thoma B. Translating targeted temperature management trials into Postarrest care. N Engl J Med 2021;384:2344–2345.

14. Callaway CW, Coppler PJ, Faro J, et al. Association of Initial Illness Severity and Outcomes after Cardiac Arrest with Targeted Temperature Management at 36°C or 33°C. JAMA Netw Open 2020;3:e208215.

15. Johnson NJ, Danielson KR, Counts CR, et al. Targeted temperature management at 33 versus 36 degrees: a retrospective cohort study. Crit Care Med 2020;48:362–369.

16. Fernando SM, Di Santo P, Sadeghirad B, et al. Targeted temperature management following out-of-hospital cardiac arrest: a systematic review and network meta-analysis of temperature targets. Intensive Care Med 2021;47:1078–1088.

17. Moseby-Knappe M, Mattsson-Carlgren N, Stammet P, et al. Serum markers of brain injury can predict good neurological outcome after out-of-hospital cardiac arrest. Intensive Care Med 2021;47:984–994.

18. Geocadin RG. Moving beyond one-size-fits-all treatment for patients after cardiac arrest. JAMA Netw Open 2020;3:e208809.

19. Elmer J, He Z, May T, et al. Precision Care in Cardiac Arrest: ICECAP (PRECICECAP) study protocol and informatics approach. Neurocrit Care 2022;37:237–247.

20. Levy DE, Caronna JJ, Singer BH, et al. Predicting outcome from hypoxic-ischemic coma. JAMA 1985;253:1420–1426.

21. Wijdicks EF, Hijdra A, Young GB, et al. Practice parameter: prediction of outcome in comatose survivors after cardiopulmonary resuscitation (an evidence-based review): report of the quality standards Subcommittee of the American Academy of neurology. Neurology 2006;67:203–210.

22. Geocadin RG, Callaway CW, Fink EL, et al. Standards for studies of neurological prognostication in comatose survivors of cardiac arrest: a scientific statement from the American Heart Association. Circulation 2019;140:e517–e542.

23. Kalanuria AA, Geocadin RG. Early prognostication in acute brain damage: where is the evidence? Curr Opin Crit Care 2013;19:113–122.

24. Sandroni C, Cariou A, Cavallaro F, et al. Prognostication in comatose survivors of cardiac arrest: an advisory statement from the European resuscitation council and the European Society of Intensive Care Medicine. Resuscitation 2014;85:1779–1789.

25. Mulder M, Gibbs HG, Smith SW, et al. Awakening and withdrawal of life-sustaining treatment in cardiac arrest survivors treated with therapeutic hypothermia*. Crit Care Med 2014;42:2493–2499.

26. Zanyk-McLean K, Sawyer KN, Paternoster R, et al. Time to awakening is often delayed in patients who receive targeted temperature management after cardiac arrest. Ther Hypothermia Temp Manag 2017;7:95–100.

27. Tsai MS, Chen WJ, Chen WT, et al. Should we prolong the observation period for neurological recovery after cardiac arrest? Crit Care Med 2022;50:389–397.

28. Greer DM, Shemie SD, Lewis A, et al. Determination of brain death/death by neurologic criteria: the world brain death project. JAMA 2020;324:1078–1097.

29. Becker LB, Aufderheide TP, Geocadin RG, et al. Primary outcomes for resuscitation science studies: a consensus statement from the American Heart Association. Circulation 2011;124:2158–2177.

30. Sawyer KN, Camp-Rogers TR, Kotini-Shah P, et al. Sudden cardiac arrest survivorship: a scientific statement from the American Heart Association. Circulation. 2020;141:e654–e685.

31. Joshi VL, Tang LH, Mikkelsen TB, et al. Does time heal fatigue, psychological, cognitive and disability problems in people who experience an out-of-hospital cardiac arrest? Results from the DANCAS survey study. Resuscitation 2023;182:109639.

32. Agarwal S, Birk JL, Abukhadra SL, et al. Psychological distress after sudden cardiac arrest and its impact on recovery. Curr Cardiol Rep 2022;24:1351–1360.

15318249, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ana.26619, Wiley Online Library on [24/04/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License